LESSON 5 · FILES AND CHECKS

Keep Scans, Masks and Results Connected

Follow one teaching participant, sub-PD01, from raw files to a measured mask

1 · RAW SCANS

raw_dicoms/sub-PD01/

```
MPRAGE_DICOM.zip
FLAIR_DICOM.zip
```

after **bash scripts/unpack_dicoms.sh**

```
MPRAGE_DICOM/ 200 .dcm
FLAIR_DICOM/ 44 .dcm
```

unpack →

2 · CONVERTED IMAGES

converted_niftis/sub-PD01/

```
sub-PD01_MPRAGE.nii.gz
sub-PD01_MPRAGE.json
sub-PD01_FLAIR.nii.gz
sub-PD01_FLAIR.json
```

from **bash scripts/convert_all_subjects_TEMPLATE.sh**, once you finish it

ITK-SNAP →

3 · YOUR MASK

lesion_masks/

```
lesion_ROI_INITIALS_sub-PD01.nii.gz
```

Example from the lesson: `lesion_ROI_SAM_sub-PD01.nii.gz`

```
label 1 = marked
label 0 = background
```

measure →

4 · RESULTS

results/

```
lesion_voxel_counts.csv
one row per subject:
sub-PD01, PD, <mask file>, <voxels>, …, ok
```

then **bash scripts/compare_groups.sh**

```
group_comparison.txt
```

GROUP LOOKUP `subject_groups.csv` `sub-PD01,PD` supplies the group in step 4

THREE SITUATIONS IN results/lesion_voxel_counts.csv THAT LOOK SIMILAR BUT ARE DIFFERENT

SITUATION	LABEL_1_VOXELS	STATUS	MASK_FILE	WHAT TO DO
Saved mask, nothing marked	`0`	`ok`	mask file name	A valid result. Keep it; it is not an error.
No mask file found	`NA`	`needs_mask`	`MISSING`	Save a mask with the subject ID in its name.
More than one mask matches	`NA`	`choose_one_mask`	`AMBIGUOUS`	Keep one mask, or enter your initials when the script asks.

Counts are marked voxels, not physical volume. Marks are annotation practice and do not indicate a diagnosis.